

Exercise Sheet 2: Sketch Solutions

Advanced Microeconomics: Economic Decision Making

Question 1

Let $z_L(w, q)$ be conditional labour demand, differentiable, h.d.0 in $w = (w_1, \dots, w_n)$, with elasticities $\varepsilon_{Lj} = \partial \ln z_L / \partial \ln w_j$ and $\varepsilon_{LL} \leq 0$.

- (a) Prove $\sum_j \varepsilon_{Lj} = 0$. [7]
- (b) All input prices change by proportions $(g_1, \dots, g_n)$. Prove, to first order,

$$d \ln z_L = \sum_{j \neq L} \varepsilon_{Lj} (g_j - g_L),$$

so that only *relative* price changes move the input mix. [10]

- (c) An economy indexes wages to a consumer price index containing import prices that never enter any domestic firm's revenues. Using (b), state the exact condition under which such indexation leaves every firm's input mix unchanged, and the sign of the labour-mix response when the condition fails. [8]

Solution. (a) Differentiate $z_L(\lambda w, q) = z_L(w, q)$ in λ at 1: $\sum_j w_j \partial z_L / \partial w_j = 0$; divide by z_L : $\sum_j \varepsilon_{Lj} = 0$.
 (b) $d \ln z_L = \sum_j \varepsilon_{Lj} g_j = \sum_j \varepsilon_{Lj} g_j - g_L \sum_j \varepsilon_{Lj}$ (subtracting zero by (a)) $= \sum_j \varepsilon_{Lj} (g_j - g_L) = \sum_{j \neq L} \varepsilon_{Lj} (g_j - g_L)$.
 (c) The mix is unchanged iff g_L equals each g_j it is compared against with nonzero elasticity — in the two-composite case, iff the wage rises exactly as fast as the firm's *other input costs*: $g_w = g_O$. If the CPI outruns firms' input costs (imported inflation), then $g_w > g_O$ and, since the own effect dominates ($\varepsilon_{LL} = -\sum_{j \neq L} \varepsilon_{Lj} \leq 0$ with cross-elasticities ≥ 0 in the two-input case), $d \ln z_L < 0$: labour is substituted away in proportion to the wedge $g_w - g_O$, whatever the level of inflation.

Question 2

A CRS competitive industry has labour cost share $s_L = 0.20$, other-input share $s_O = 0.80$, substitution elasticity $\sigma = 0.5$, and faces industry demand of elasticity $\varepsilon_D = -0.5$. In a shock year, other input costs rise $g_O = 8\%$; indexation raises wages $g_w = 12\%$.

- (a) Compute, to first order, the output price, output, and total labour responses for the cost shock alone ($g_w = 0$). [9]
- (b) Repeat with the indexation added, and compute the policy's marginal contribution to each. [10]

- (c) Compute the labour-share-of-revenue response in both runs and give a one-sentence reading of the policy's marginal effect on it. [6]

Solution. Use $d \ln p = s_O g_O + s_L g_w$; $d \ln q = \varepsilon_D d \ln p$; $d \ln z_L|_q = -\sigma(1 - s_L)(g_w - g_O)$; $d \ln L = d \ln z_L|_q + d \ln q$; $d \ln LS = g_w + d \ln L - d \ln p - d \ln q$.

(a) $d \ln p = 0.8 \times 8 = 6.4\%$; $d \ln q = -3.2\%$; mix: $-0.5 \times 0.8 \times (0 - 8) = +3.2\%$ (labour is the refuge); $d \ln L = 3.2 - 3.2 = 0$.

(b) $d \ln p = 6.4 + 0.2 \times 12 = 8.8\%$; $d \ln q = -4.4\%$; mix: $-0.5 \times 0.8 \times (12 - 8) = -1.6\%$; $d \ln L = -1.6 - 4.4 = -6.0\%$. Policy marginal: $\Delta p : +2.4\text{pp}$; $\Delta q : -1.2\text{pp} \times 2 = -2.4\text{pp}$; $\Delta L : -6.0\text{pp}$.

(c) Run 1: $d \ln LS = 0 + 0 - 6.4 + 3.2 = -3.2\%$. Run 2: $12 - 6.0 - 8.8 + 4.4 = +1.6\%$. Policy marginal = +4.8%: indexation flips the labour share from falling to rising — it protects labour's slice of income while shrinking employment, the insider-marginal-worker split in one number.

Question 3

Technology $q = z_1^\alpha z_2^{1-\alpha}$, $\alpha \in (0, 1)$.

- (a) Solve the CMP: derive $z(w, q)$ and show $c(w, q) = q \kappa w_1^\alpha w_2^{1-\alpha}$ with $\kappa = \alpha^{-\alpha}(1 - \alpha)^{-(1-\alpha)}$. [12]
- (b) Verify Shephard's lemma, and show the cost shares equal the exponents. [7]
- (c) Input prices change by proportions $(g_1, g_2) = (12\%, 3\%)$ with $\alpha = 0.25$. Compute $d \ln c$ to first order and interpret each term. [6]

Solution. (a) FOC ratio: $\frac{\alpha z_2}{(1 - \alpha) z_1} = \frac{w_1}{w_2}$, so $z_2 = \frac{(1 - \alpha) w_1}{\alpha w_2} z_1$. Impose $q: z_1^\alpha z_2^{1-\alpha} = q \Rightarrow z_1 = q \left(\frac{\alpha w_2}{(1 - \alpha) w_1} \right)^{1-\alpha}$, $z_2 = q \left(\frac{(1 - \alpha) w_1}{\alpha w_2} \right)^\alpha$. Cost: $w_1 z_1 + w_2 z_2 = q w_1^\alpha w_2^{1-\alpha} [\alpha^{-\alpha}(1 - \alpha)^\alpha \cdot \frac{\alpha}{\alpha} \dots]$; collecting powers gives $c = q \kappa w_1^\alpha w_2^{1-\alpha}$.

(b) $\partial c / \partial w_1 = \alpha c / w_1$; and from (a), $z_1 = \alpha c / w_1$ (compute $w_1 z_1 = \alpha c$ directly): Shephard holds, and $s_1 = w_1 z_1 / c = \alpha$.

(c) $d \ln c = s_1 g_1 + s_2 g_2 = 0.25 \times 12 + 0.75 \times 3 = 3 + 2.25 = 5.25\%$: each term is exposure $\times$ shock — the firm's Laspeyres, exact to first order by Shephard (re-optimisation is second order).